

NATIONAL REAL-TIME PAYMENTS SYSTEM

Technical Architecture & Implementation Report

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1. Introduction

This report presents the technical architecture for a National Real-Time Payments System (NRPS) for India, serving 150 million users with 50,000+ transactions per second at peak. The architecture follows the Central Switch pattern employed by India's UPI, Brazil's PIX, the UK's Faster Payments, and the US's FedNow.

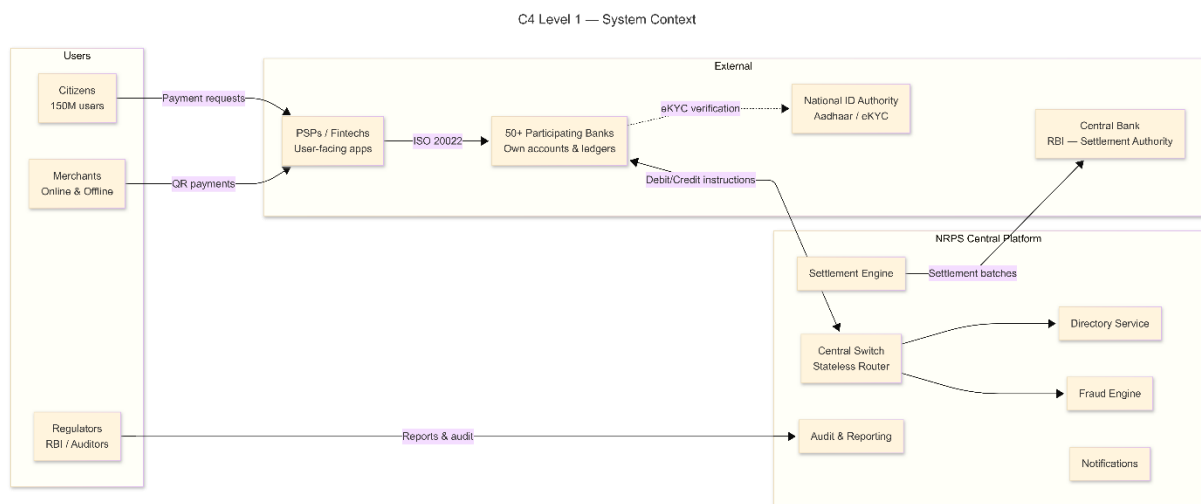
The document covers domain analysis, functional and non-functional requirements, capacity planning, system design with C4 model diagrams, technology decisions, cost estimation, a 24-month implementation roadmap, and architecture decision records.

2. Domain Understanding

2.1 The Central Switch Pattern

Every successful national real-time payment system shares three structural characteristics. First, a central stateless switch routes payment messages between participating banks without storing account balances or processing transactions itself. Second, banks manage their own accounts, balances, KYC, and transaction processing on their own infrastructure. Third, settlement between banks is deferred into batch cycles (UPI uses 10-12 daily cycles) whilst users see instant payment confirmation.

This pattern distributes the workload across the banking system. The switch handles 50,000+ messages per second, but the ACID transactional work is shared across 50 or more participating banks, each handling a manageable fraction.



2.2 Global Benchmarks

System	Country	Users	Monthly Txns	Pattern	Operator
UPI	India	500M+	20 billion	Central Switch	NPCI
PIX	Brazil	175M	Billions	Central Switch	Central Bank
Faster Payments	UK	~53M adults	Hundreds of millions	Central Switch	VocaLink

FedNow	USA	1,300+ FIs	Growing	Central Switch (RTGS)	Federal Reserve
NRPS (ours)	India	150M	~5 billion	Central Switch	-

2.3 Responsibility Boundaries

Central platform (our scope): payment switch (message routing), directory service (alias-to-bank mapping), settlement engine (net position calculation), fraud scoring (central ML), transaction logging, regulatory reporting, participant certification, rule-setting, and ISO 20022 API specification.

Participating banks: customer onboarding and Aadhaar eKYC, account management, balance tracking, debit and credit processing, bank-level fraud checks, user authentication (device binding, mPIN, biometric), transaction history, and integration adapter development.

Payment Service Providers (PSPs): user-facing mobile and web applications, QR code generation and scanning, payment initiation. PSPs connect via their sponsor bank, not directly to the switch.

3. Assumptions and Capacity Estimations

Parameter	Value	Basis
Transaction payload	1-2 KB	ISO 20022 message standard
Txns per active user per day	4	Benchmark against UPI
Peak-to-average ratio	10x	Festival and salary day spikes
Participating banks	50+	Major banks mandatory, others optional
Data retention	7 years	Regulatory requirement (RBI)
Settlement cycles	Multiple daily (configurable)	UPI uses 10-12 cycles/day
Deployment	3 national DCs, active-active switch	Five-nines availability target

Metric	Normal	Peak (10x)	Notes
Switch msg/sec	50,000	500,000	Stateless routing, not ACID writes
Per-bank avg TPS	~1,000	~10,000	50 banks share ACID workload
Daily transactions	160M	1.6B	40M DAU × 4 transactions
Transaction log/day	320 GB	3.2 TB	Append-only to Kafka + Cassandra
Directory size	~30 GB	-	150M users × 200 bytes; fits in Redis
Annual storage	~140 TB	-	Including logs, audit, replicas
Switch compute	Scales horizontally	2014	Stateless; node count determined during detailed sizing

Storage follows tiered archival: hot (90 days, SSD), warm (2 years, block storage), and cold (7 years, object storage), reducing annual cost by approximately 60%.

4. Functional Requirements

4.1 Payment Routing

The switch accepts ISO 20022 payment instructions from banks, resolves the recipient's bank via the directory, routes debit and credit instructions, collects responses, and returns final status. End-to-end within 2 seconds, with the switch's contribution under 50 milliseconds.

4.2 Directory Service

Maps payment aliases (phone numbers, Aadhaar-linked IDs, VPAs, random keys) to bank account details. Supports registration, deregistration, and transfer between banks. Sub-millisecond lookups from the Redis cache layer.

4.3 Settlement

Computes net positions between all participating banks in configurable batch cycles (comparable to UPI's 10-12 daily cycles). Generates settlement instructions for the central bank (RBI). Supports full reconciliation against the transaction log.

4.4 Fraud Detection

Every transaction is scored against an ML model asynchronously via Kafka. Transactions exceeding a configurable risk threshold (default 0.85) are flagged. Normal-risk transactions proceed without delay. Complements bank-level fraud detection.

4.5 Audit and Regulatory Reporting

Immutable, append-only transaction log retained for 7 years per RBI requirements. Supports scheduled and ad-hoc regulatory queries via Elasticsearch. Aggregate reports by geography, user type, transaction type, and bank.

4.6 Participant Management

Onboards and certifies banks and PSPs. Publishes API specifications and SDKs. Monitors compliance and enforces rules. Manages inter-participant dispute resolution.

5. Non-Functional Requirements

5.1 Performance

The switch must route 500,000 messages per second at peak. Since it is stateless, scaling is achieved by adding switch instances on Kubernetes. End-to-end latency must not exceed 2 seconds at the 99th percentile.

5.2 Availability

Target: 99.999% (5.26 minutes downtime per year). Achieved through active-active deployment of the stateless switch across three national data centres with DNS-based failover. RTO for DC-level failover under 30 seconds.

5.3 Security and Compliance

PCI DSS Level 1 and ISO 27001 certification before go-live. mTLS for all bank-to-switch communication. AES-256 at rest with HSM-managed keys. ISO 20022 mandatory message standard. All three data centres within India.

5.4 Authentication Model

Authentication is enforced at the bank/PSP layer (not the central switch) and follows a four-layer model aligned with RBI guidelines and the UPI authentication framework:

Layer 1 - Device Binding: Transaction can only be initiated from a registered device. SIM and device fingerprint are bound during onboarding.

Layer 2 - User PIN (mPIN): A user-set 4-6 digits PIN is required for every transaction. The PIN is encrypted end-to-end and validated by the issuing bank.

Layer 3 - Biometric (Optional): Fingerprint or face recognition, enforced by the bank or PSP based on risk policy.

Layer 4 - OTP for High-Value: One-time passcode via SMS for transactions exceeding a configurable threshold, providing an additional verification layer.

5.5 Reliability

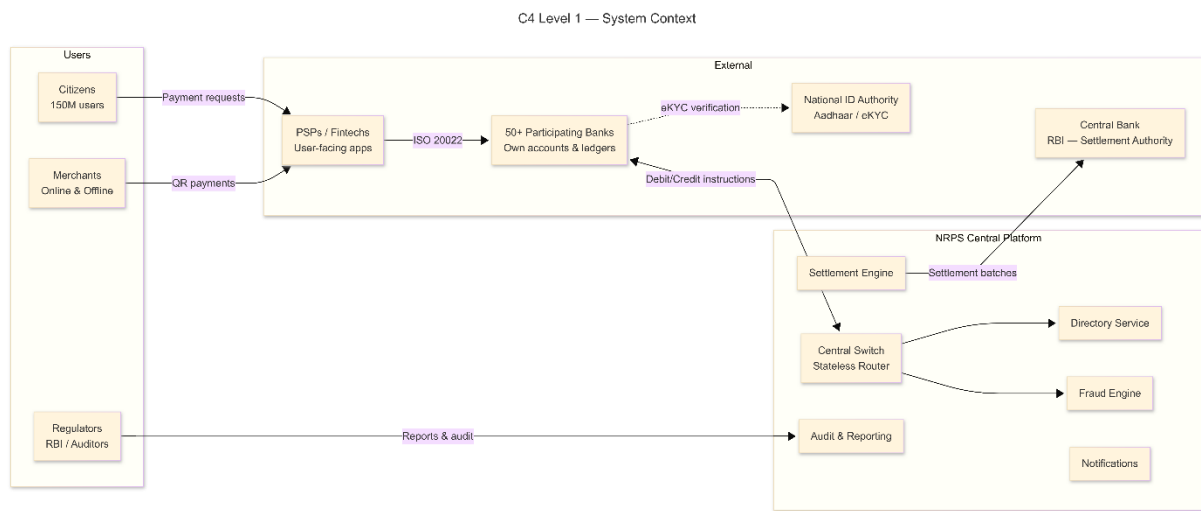
Per-bank circuit breakers isolate failing banks. Transaction timeout of 30 seconds triggers auto-reversal. RPO within a DC is zero (synchronous replication). Graceful degradation ensures one bank's outage does not affect others.

6. System Design

The architecture follows the Central Switch pattern verified against UPI, PIX, Faster Payments, and FedNow. Four C4 diagrams are provided as accompanying Mermaid files.

6.1 System Context (C4 Level 1)

The NRPS Central Platform sits between users (citizens, merchants, regulators including RBI) and external systems (50+ participating banks, the central bank, PSPs, and the Aadhaar/eKYC identity authority). The platform owns the switch, directory, settlement, fraud scoring, and reporting. Banks and PSPs operate outside this boundary.



6.2 Containers (C4 Level 2)

API Gateway (Kong + NGINX): Accepts ISO 20022 messages over mTLS. Rate limiting and schema validation.

Central Switch (Go): Stateless routing engine. Resolves recipient bank from directory (Redis), routes debit to sender's bank, waits for confirmation, routes credit to receiver's bank, returns status. Publishes events to Kafka. Per-bank circuit breakers. Scales horizontally.

Directory Service (Redis Cluster + PostgreSQL): Full dataset (~30 GB) in Redis for O(1) lookups. PostgreSQL for durable backing store.

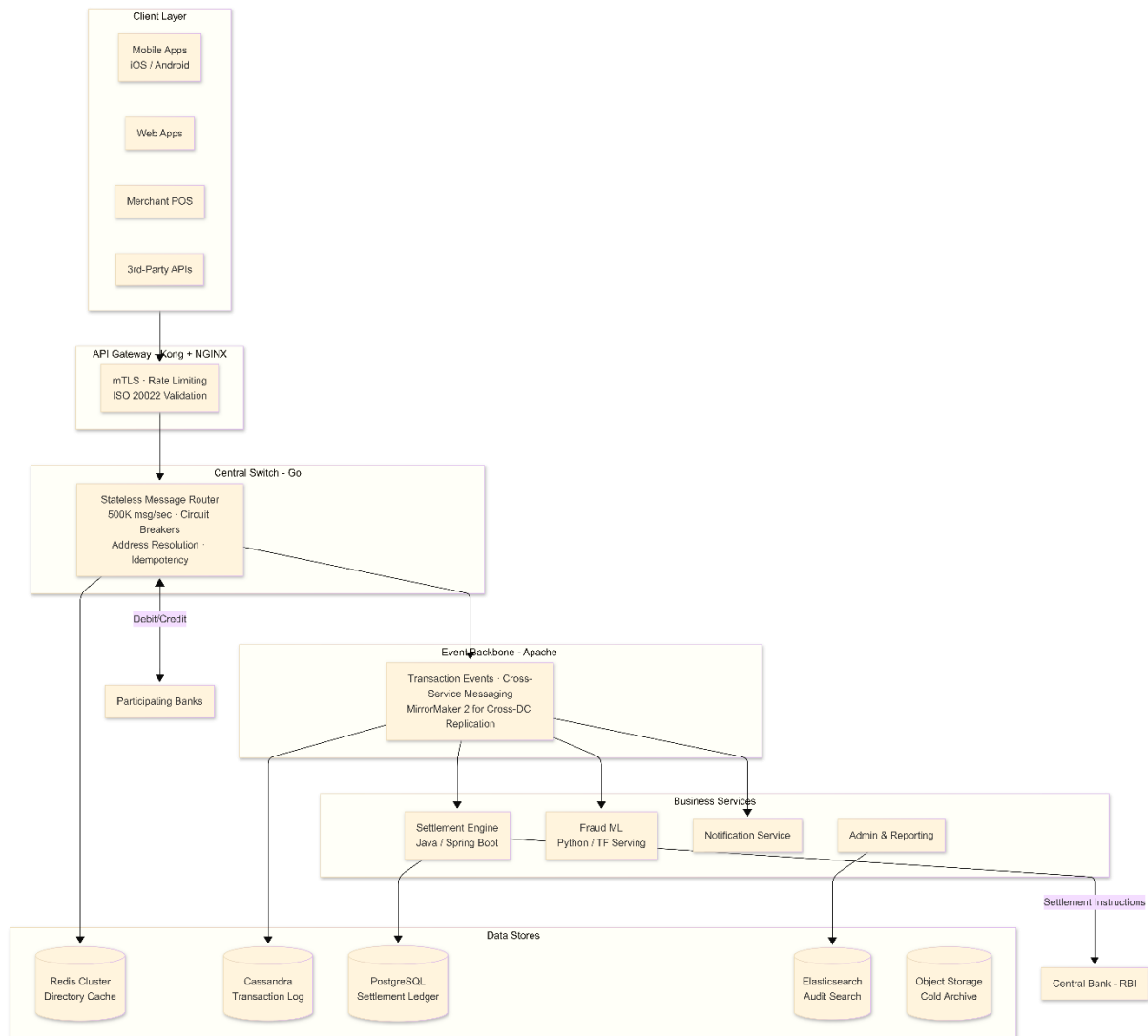
Settlement Engine (Java/Spring Boot + PostgreSQL): Consumes events from Kafka, computes net positions in batch cycles, generates settlement instructions for RBI. Standard PostgreSQL with Patroni HA.

Fraud Detection (Python + TensorFlow Serving): Async scoring via Kafka in under 100ms.

Event Backbone (Apache Kafka): Replication factor 3 per DC. MirrorMaker 2 for cross-DC.

Data Stores: Cassandra (append-only transaction log, multi-DC). Elasticsearch (audit search). S3 (cold archival, KYC documents).

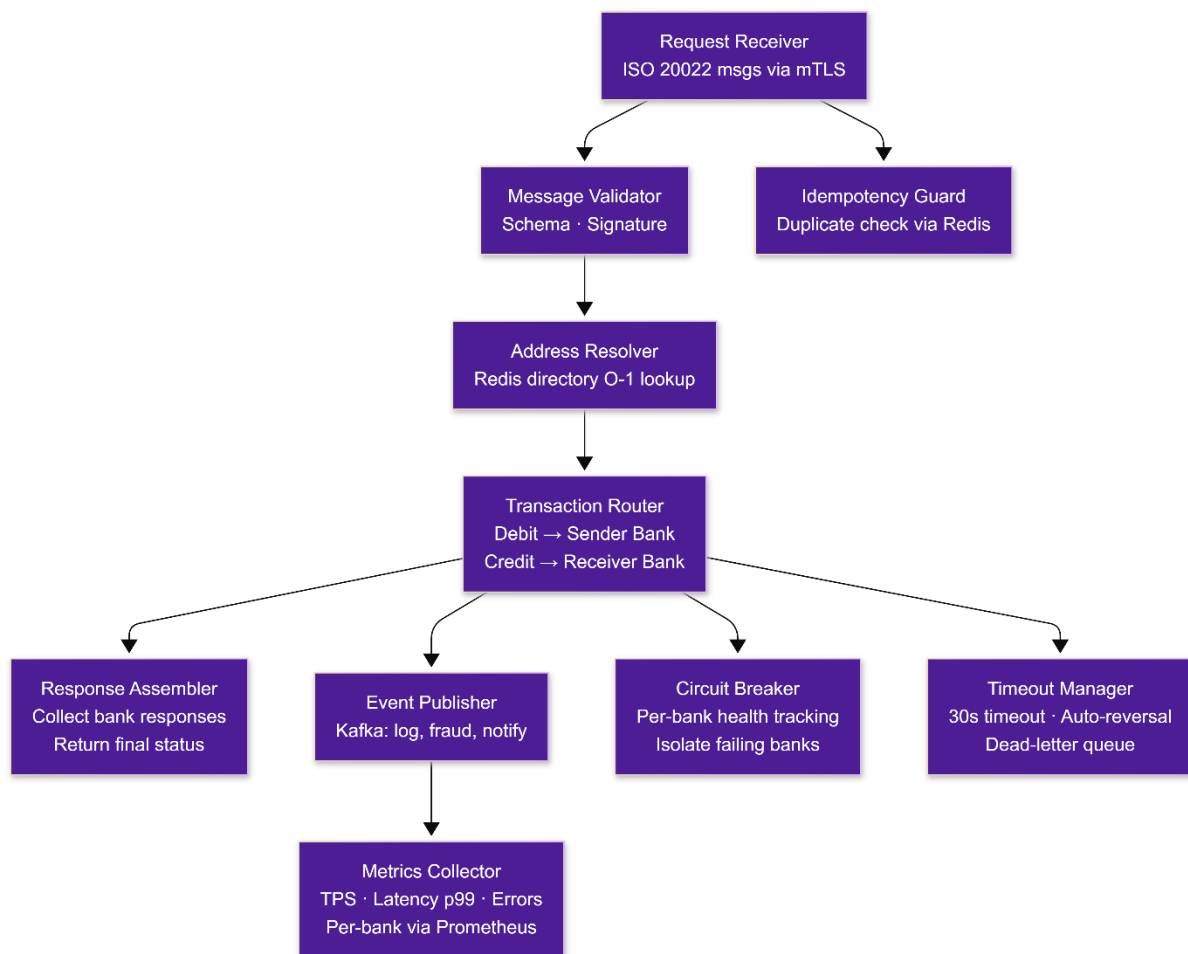
C4 Level 2 — Container Diagram



6.3 Central Switch Components (C4 Level 3)

Inside the switch, ten stateless components handle the message flow: Request Receiver (ISO 20022 ingestion), Message Validator (schema and signature verification), Idempotency Guard (duplicate detection via Redis), Address Resolver (recipient bank lookup), Transaction Router (debit/credit instruction dispatch), Event Publisher (Kafka event emission), Response Assembler (bank response aggregation), Circuit Breaker (per-bank health isolation), Timeout Manager (30-second auto-reversal), and Metrics Collector (Prometheus instrumentation). No component performs ACID database operations in the hot path.

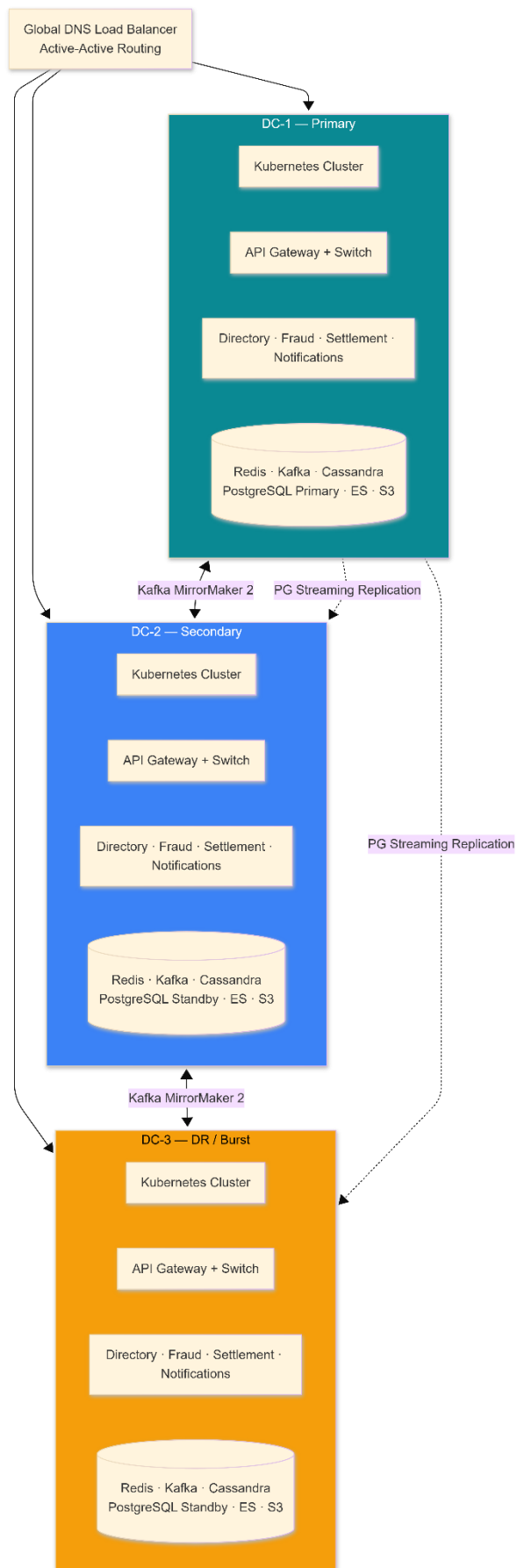
C4 Level 3 — Central Switch Components



6.4 Deployment Topology (C4 Level 4)

Three national data centres each run the full stack on Kubernetes. The stateless switch is active-active across all three. PostgreSQL uses streaming replication (primary in DC-1, standbys in DC-2 and DC-3). Kafka replicates across DCs via MirrorMaker 2. Cassandra replicates natively. Global DNS routes traffic to the nearest healthy DC.

C4 Level 4 — Deployment Topology



6.5 Transaction Flow

A user initiates a payment through their PSP's app, authenticating via device binding and mPIN. The PSP sends the instruction to its bank, which forwards it to the central switch in ISO 20022 format. The switch validates, deduplicates, resolves the recipient's bank, and routes a debit instruction to the sender's bank.

The sender's bank verifies the account, checks the balance, performs its own fraud checks, debits, and confirms. The switch routes a credit instruction to the receiver's bank, which credits and confirms. Both parties are notified. The event is logged to Kafka and persisted to Cassandra. Net positions settle in the next batch cycle.

If credit fails after debit, the switch instructs reversal. Stuck transactions auto-reverse after 30 seconds.

7. Technology Decisions

Component	Technology	Rationale
API Gateway	Kong + NGINX	mTLS, rate limiting, ISO 20022 validation
Central Switch	Go (recommended)	Goroutine model, stateless scaling. Note: UPI uses Java/Spring.
Directory	Redis Cluster + PostgreSQL	30 GB in memory, O(1) lookups, PG for durability
Transaction Log	Kafka + Cassandra	Append-only, multi-DC, exactly-once semantics
Settlement	Java/Spring Boot + PostgreSQL	Batch ACID, banking integration ecosystem
Fraud	Python + TensorFlow Serving	Async scoring <100ms, model versioning
Audit	Elasticsearch	Full-text regulatory queries at scale
Monitoring	Prometheus + Grafana + Jaeger	Metrics, alerting, distributed tracing
Deployment	Kubernetes (3 national DCs)	Auto-scaling, rolling deploys, active-active switch

Every technology is open-source and proven in production at scale beyond NRPS requirements.

8. Cost and Implementation

8.1 Cost Reference 0026 Indicative Estimate

NPCI2019s published financials (FY 2024201325) show that India2019s entire payment switching infrastructure (UPI + IMPS + NACH) operates at a total tech and depreciation cost of approximately 20B9500 Crore (~\$59M) per year, on a cumulative asset block of 20B91,560 Crore (~\$184M). NPCI generates 20B94,000 Crore in revenue with EBITDA margins above 50%, processing over 20 billion transactions monthly. This provides the most relevant cost benchmark for the NRPS. Bank-side integration is each bank's responsibility.

Category	Year 1 (Build + Run)	Annual (Year 2+)
Infrastructure (3 DCs)	\$8-12M	\$5-8M
Software and tooling	\$1-2M	\$1-1.5M
Engineering (40-60 people)	\$8-12M	\$6-9M
Security, compliance, audit	\$3-5M	\$2-3M
Testing and performance	\$2-3M	\$1-2M
Total	\$22-34M	\$15-23.5M

8.2 Roadmap

Phase 1 (Months 1-6): Switch MVP, directory service, ISO 20022 API spec, top 10 bank integrations, HSM + mTLS, single-DC deployment.

Phase 2 (Months 7-12): Remaining bank onboarding, QR/P2M payments, fraud detection ML v1, settlement automation, regulatory dashboards, second DC.

Phase 3 (Months 13-18): Full HA across 3 DCs, 10x spike validation, PCI DSS and ISO 27001 certification, dispute management, participant certification.

Phase 4 (Months 19-24): Cross-border preparation, currency conversion, international partner APIs, multi-language support, production handover.

9. Architecture Decision Records

ADR-001: Central Switch with Federated Banks

Context: National real-time payment systems require a scalable, interoperable architecture. A monolithic approach would centralise all transaction processing.

Decision: Adopt the Central Switch pattern: a stateless message router with banks handling their own accounts and transactions.

Trade-offs: The switch cannot enforce balance checks directly. Misbehaving banks are detected through reconciliation and regulatory enforcement.

ADR-002: Deferred Net Settlement (Batch Cycles)

Context: Users expect instant confirmation, but real-time gross settlement for every transaction at 500K TPS is impractical.

Decision: Instant user confirmation via the switch's transaction log. Inter-bank settlement in configurable batch cycles (comparable to UPI's 10-12 daily cycles).

Trade-offs: Banks carry counterparty risk between transaction and settlement. Standard and accepted across UPI, PIX, and Faster Payments.

ADR-003: Multi-Factor Authentication (4 Layers)

Context: Payment security requires strong authentication without impacting transaction speed.

Decision: Four-layer model: device binding, user mPIN (every transaction), optional biometric, and OTP for high-value. Aligned with RBI guidelines and UPI's authentication framework.

Trade-offs: Authentication is enforced at the bank/PSP layer, not the central switch. Requires all participants to implement consistently.

ADR-004: Go for Switch, Java for Business Services

Context: The switch requires raw throughput. Business services require rich banking integrations.

Decision: Go for the stateless switch (goroutine concurrency). Java/Spring Boot for settlement, fraud orchestration, and reporting. Note: UPI uses Java/Spring for its switch; Go is our recommendation for raw routing throughput.

Trade-offs: Two-language codebase. Mitigated by clear service boundaries.

ADR-005: Kafka + Cassandra for Transaction Log

Context: The log is append-only, write-heavy, and requires multi-DC replication. No SQL queries in the hot path.

Decision: Kafka for event streaming. Cassandra for durable multi-DC storage. Elasticsearch for queryable audit.

Trade-offs: Cassandra's eventual consistency is acceptable for an append-only log.

ADR-006: ISO 20022 Mandatory Message Standard

Context: Interoperability between diverse bank systems requires standardisation. ISO 20022 is adopted by SWIFT, SEPA, and the UK's NPA.

Decision: Mandate ISO 20022 for all switch communication. Publish specifications and a certification programme.

Trade-offs: Banks must build or adapt ISO 20022 adapters. Standard industry expectation.

10. Conclusion

The architecture follows the Central Switch pattern proven at national scale by UPI, PIX, Faster Payments, and FedNow. By distributing responsibilities between a stateless routing engine and participating banks, it achieves the required scale (500K messages per second), reliability (99.999%), and latency (sub-2-second) using established, open-source technology.

The estimated investment of \$22-34 million for Year 1, reducing to \$15-23.5 million annually, reflects the focused scope of the central platform. The phased implementation delivers a working payment flow within six months and full production capability within 24 months.

We recommend proceeding with stakeholder alignment, followed by an 8-week proof of concept with two participating banks.

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